

Assignment II

(Series Solution and Special Function)

Subject: Calculus-II (MTH 210)

Deadline: 2080-09-01

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1. Solve by power series method:

(i) $(1-x)y' = y$ **Ans:** $y = \frac{a_o}{1-x}$

(ii) $xy' - 3y = k$, where k is constant. **Ans:** $y = a_3x^3 - \frac{k}{3}$

(iii) $y'' - 4y = 0$ **Ans:** $y = a_o(1 + 2x^2 + \frac{2x^4}{3} + \dots) + a_1(x + \frac{2}{3}x^3 + \frac{2x^5}{15} + \dots)$

(iv) $y'' - 4xy' + (4x^2 - 2)y = 0$ **Ans:** $y = a_o(1 + x^2 + \frac{x^4}{2} + \dots) + a_1(x + x^3 + \frac{x^5}{2} + \dots)$

2. Define Legendre's equation. Also derive the solution of Legendre's equation.

3. What is Legendre's equation? Find Legendre's polynomial.

4. Define Bessel's differential equation and find the Bessel's function of first kind.

5. Show that: $P_n(-x) = (-1)^n P_n(x)$.

NOTE: late submission will not consider for evaluation.